

Abstract

These notes derive the Klein-Gordon equation starting from the relativistic energy-momentum relation. By promoting the classical energy and momentum to operators via the substitutions $E \rightarrow i\hbar \frac{\partial}{\partial t}$ and $p \rightarrow -i\hbar \nabla$, we obtain a wave equation for a scalar field ψ . Our goal is to present the covariant form of the equation and explain its Lorentz covariance for scalar fields.

The Klein-Gordon equation in manifestly covariant form is

$$\left(\square - \frac{m^2 c^2}{\hbar^2} \right) \psi = 0 \quad (1)$$

Here, $\square = \partial_\mu \partial^\mu$ is the d'Alembert operator. It is often convenient to write the equation in the equivalent form

$$\square \psi = \mu^2 \psi, \quad \mu = \frac{mc}{\hbar} \quad (2)$$

In these notes, we will work in covariant formalism to make Lorentz covariance transparent and to clarify the transformation properties of scalar fields.

Derivation from the energy-momentum relation

We start from the relativistic energy-momentum relation.

$$E^2 = p^2 c^2 + m^2 c^4 \quad (3)$$

To introduce quantization, we promote the energy and momentum to operators.

$$E \rightarrow i\hbar \frac{\partial}{\partial t}, \quad p \rightarrow -i\hbar \nabla \quad (4)$$

These operators act on a scalar wavefunction ψ .

$$\left(i\hbar \frac{\partial}{\partial t} \right)^2 \psi = \left[(-i\hbar \nabla)^2 c^2 + m^2 c^4 \right] \psi \quad (5)$$

Expanding the operators gives the following differential equation.

$$-\hbar^2 \frac{\partial^2 \psi}{\partial t^2} = -\hbar^2 c^2 \nabla^2 \psi + m^2 c^4 \psi \quad (6)$$

Rearranging terms, we obtain the non-covariant form.

$$\left[\hbar^2 c^2 \left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) - m^2 c^4 \right] \psi = 0 \quad (7)$$

This is the Klein-Gordon equation before introducing covariant notation.

Covariant formalism

We now introduce the Minkowski metric convention $\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1)$ and the d'Alembert operator.

$$\partial_\mu \partial^\mu = \square = -\frac{1}{c^2} \frac{\partial^2}{\partial t^2} + \nabla^2 \quad (8)$$

The non-covariant result can now be rewritten more compactly.

$$\left[\hbar^2 c^2 \square - m^2 c^4 \right] \psi = 0 \quad (9)$$

It is convenient to introduce the parameter μ ,

$$\mu = \frac{mc}{\hbar} \quad (10)$$

Division by $\hbar^2 c^2$ then gives the standard covariant form.

$$\left(\square - \mu^2 \right) \psi = 0 \quad (11)$$

This may be written equivalently as

$$\square \psi = \mu^2 \psi \quad (12)$$

Here, the four-gradient and four-position are defined as

$$\partial_\mu = \left(c^{-1} \partial_t, \nabla \right), \quad x^\mu = (ct, x, y, z) \quad (13)$$

Note that μ has dimensions of inverse length and equals the inverse reduced Compton wavelength.

Lorentz invariance

We assume that the Klein-Gordon field is a scalar field and that spacetime coordinates transform under a Lorentz transformation according to

$$x'^{\mu} = \Lambda^{\mu}_{\nu} x^{\nu} \quad (14)$$

A scalar field transforms by composition with the inverse transformation.

$$\psi'(x') = \psi(x) = \psi(\Lambda^{-1} x') \quad (15)$$

The operator $\partial_{\mu} \partial^{\mu}$ is a Lorentz scalar operator.

$$\square' \psi'(x') = \square \psi(x) \quad (16)$$

Since μ is a scalar constant, the equation keeps the same form in all inertial frames.

$$(\square' - \mu^2) \psi'(x') = 0 \quad \text{iff} \quad (\square - \mu^2) \psi(x) = 0 \quad (17)$$

Thus, ψ should be regarded as a Lorentz scalar field rather than "invariant" in a naive sense.

Therefore, the Klein-Gordon equation is Lorentz covariant for scalar fields.

Remarks and notes

1. The metric convention used throughout these notes is

$$\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1) \quad (18)$$

With the opposite signature $(+, -, -, -)$ the sign of the d'Alembertian changes but the physical content does not.

For practical calculations it is often useful to write the Klein-Gordon equation in its expanded standard form

$$\left(-\frac{1}{c^2} \frac{\partial^2}{\partial t^2} + \nabla^2 - \frac{m^2 c^2}{\hbar^2} \right) \psi = 0 \quad (19)$$

The Klein-Gordon equation is second order in both time and space.

It describes free relativistic spin-0 particles.

The field ψ may be real or complex, depending on the physical problem.

Starting from $E^2 = p^2 c^2 + m^2 c^4$ and using the operator substitutions, one obtains a Lorentz-covariant wave equation for scalar fields.

References

The references below provide complementary perspectives on the Klein-Gordon equation and relativistic field theory. Greiner [1] introduces the relativistic wave-equation approach, Gross [2] connects it to field theory, Peskin and Schroeder [3] present the standard field-theoretic viewpoint, Srednicki [4] offers a concise modern treatment, and Weinberg [5] gives the most rigorous foundations. Together, these texts support the covariant interpretation adopted in these notes.

- [1] W. Greiner, *Relativistic Quantum Mechanics: Wave Equations*, 3rd ed., Springer, Berlin, 2000.
- [2] F. Gross, *Relativistic Quantum Mechanics and Field Theory*, John Wiley & Sons, New York, 1993.
- [3] M. E. Peskin and D. V. Schroeder, *An Introduction to Quantum Field Theory*, Addison-Wesley, Reading, Massachusetts, 1995.
- [4] M. Srednicki, *Quantum Field Theory*, Cambridge University Press, Cambridge, 2007.
- [5] S. Weinberg, *The Quantum Theory of Fields, Volume I: Foundations*, Cambridge University Press, Cambridge, 1995.